

# Next-generation Bayesian local robustness

---

Ryan Giordano ([rgiordano@berkeley.edu](mailto:rgiordano@berkeley.edu), UC Berkeley)

**Berkeley Neyman Seminar**

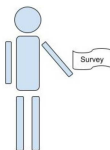
**September 2026**

# The basic problem

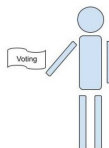
We have a survey population, for whom we observe:

- Covariates  $x$  (e.g. race, gender, zip code, age, education level)
- Responses  $y$  (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe  $(x_i, y_i)$  for  $i = 1, \dots, N_S$



Observe  $x_j$  for  $j = 1, \dots, N_T$

# The basic problem

We have a survey population, for whom we observe:

- Covariates  $x$  (e.g. race, gender, zip code, age, education level)
- Responses  $y$  (e.g. A binary response to “do you support candidate Z”)

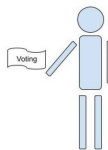
We want the average response in a target population, in which we observe only covariates.

**The problem is that the populations may be very different, maybe leading to bias.**



Photo copyright Mark Taylor: naturepl.com

Observe  $(x_i, y_i)$  for  $i = 1, \dots, N_S$



Observe  $x_j$  for  $j = 1, \dots, N_T$

# The basic problem

We have a survey population, for whom we observe:

- Covariates  $x$  (e.g. race, gender, zip code, age, education level)
- Responses  $y$  (e.g. A binary response to “do you support candidate Z”)

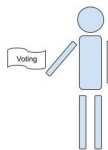
We want the average response in a target population, in which we observe only covariates.

**The problem is that the populations may be very different, maybe leading to bias.**



Photo copyright Mark Taylor: naturepl.com

Observe  $(x_i, y_i)$  for  $i = 1, \dots, N_S$



Observe  $x_j$  for  $j = 1, \dots, N_T$

This talk will study “Multilevel regression and poststratification” estimators:

- Set  $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$  and prior  $\mathcal{P}(\theta)$  (e.g. Hierarchical logistic regression)
- Estimate the posterior  $\mathcal{P}(\theta|\text{Data})$  with Markov Chain Monte Carlo (MCMC)
- Take  $\hat{\mu}^{\text{MrP}}(Y_S) = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$  where  $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\text{Data})}[y|x_j]$ .

# The basic problem

We have a survey population, for whom we observe:

- Covariates  $x$  (e.g. race, gender, zip code, age, education level)
- Responses  $y$  (e.g. A binary response to “do you support candidate Z”)

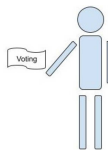
We want the average response in a target population, in which we observe only covariates.

**The problem is that the populations may be very different, maybe leading to bias.**



Photo copyright Mark Taylor: naturepl.com

Observe  $(x_i, y_i)$  for  $i = 1, \dots, N_S$



Observe  $x_j$  for  $j = 1, \dots, N_T$

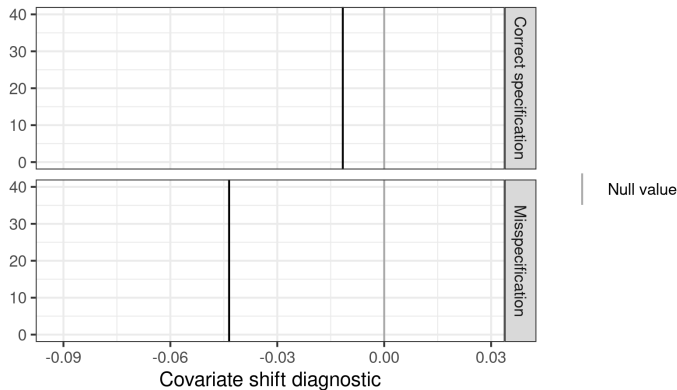
This talk will study “Multilevel regression and poststratification” estimators:

- Set  $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$  and prior  $\mathcal{P}(\theta)$  (e.g. Hierarchical logistic regression)
- Estimate the posterior  $\mathcal{P}(\theta|\text{Data})$  with Markov Chain Monte Carlo (MCMC)
- Take  $\hat{\mu}^{\text{MrP}}(Y_S) = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$  where  $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\text{Data})}[y|x_j]$ .

**Problem:** MCMC is time consuming and the regression model is surely misspecified.

**Goal:** Meaningful regression specification checks with frequentist uncertainty.

## Result preview



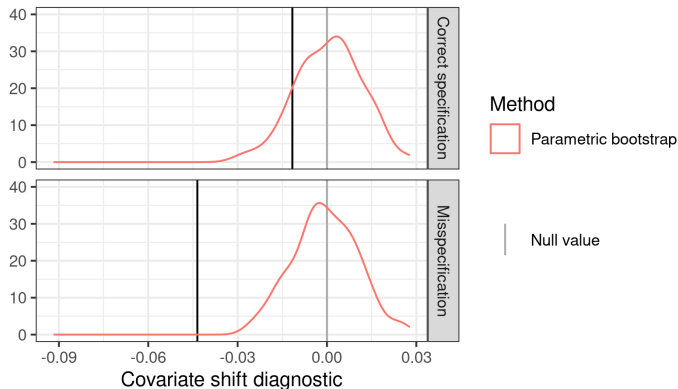
**Figure 1:** A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

**Problem:** What is “large enough” to be real evidence of misspecification?

**Possible answer:** Frequentist confidence intervals via the parametric bootstrap.

## Result preview



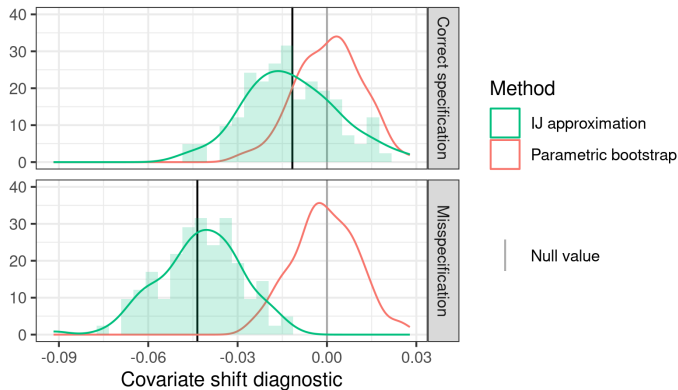
**Figure 1:** A covariate shift diagnostic

The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

**Problem:** Expensive: each bootstrap sample required a new MCMC run.

**Possible answer:** Infinitesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

## Result preview

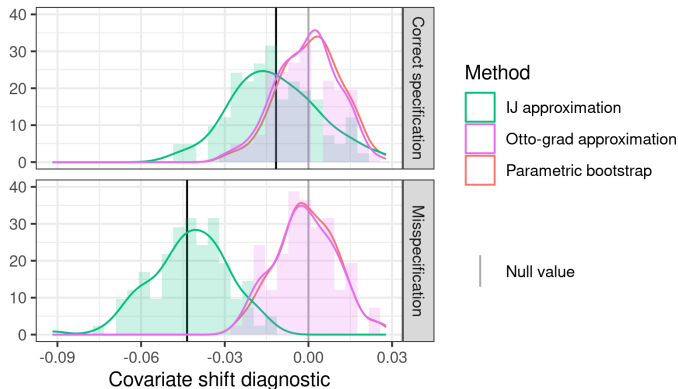


**Figure 1:** A covariate shift diagnostic

IJ confidence intervals are fast and often work well for posterior means.  
But they are not accurate for this more complex diagnostic (high Monte Carlo error).



## Result preview



**Figure 1:** A covariate shift diagnostic

IJ confidence intervals are fast and often work well for posterior means. But they are not accurate for this more complex diagnostic (high Monte Carlo error).

**Answer:** Our flow-based local sensitivity measures (“Otto grad”) produce good approximations to the bootstrap at negligible computational expense.

- Bayesian local regression specification checks
  - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
  - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
  - (With Lucas Schwengber)

# Bayesian local regression specification checks

## Local specification checks

$$\hat{\mu}^{\text{MrP}}(Y_S) = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j \text{ where } \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\text{Data})} [m(\theta^\top x_j)]}_{\text{Want to check without re-running MCMC}}.$$

This model of *survey data* is definitely misspecified.

Why do we care about correct specification? We want to use  $\mathcal{P}_S(x)$  to learn about  $\mathcal{P}_T(x)$ .

Correct specification  $\Rightarrow$  Regression estimates are invariant to  $\mathcal{P}_S(x)$

**Idea:** Check invariance to  $\mathcal{P}_S(x)$  directly.

Let  $\mathcal{P}(x, y)$  denote a generic joint density of  $x, y$ .

Let  $\hat{\mathcal{P}}_S$  the empirical distribution on the survey data.

1. Write our estimand as a functional  $\mu(\mathcal{P})$  that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation  $w(x)$
3. Define  $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x) \mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x)) \mathcal{P}(y|x)$
4. Compute the derivative  $\Delta(w, \mathcal{P}) = \partial \mu(\mathcal{P}_\epsilon) / \partial \epsilon$
5. Plug the empirical distribution into  $\Delta(w, \hat{\mathcal{P}}_S)$  and check whether  $\approx 0$

# Bayesian sensitivity analysis

Consider a family of distributions  $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$  and a test function  $\phi(\theta)$ . Then, by interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

---

Let the regression log likelihood be  $\ell(y|x, \theta)$ , so that

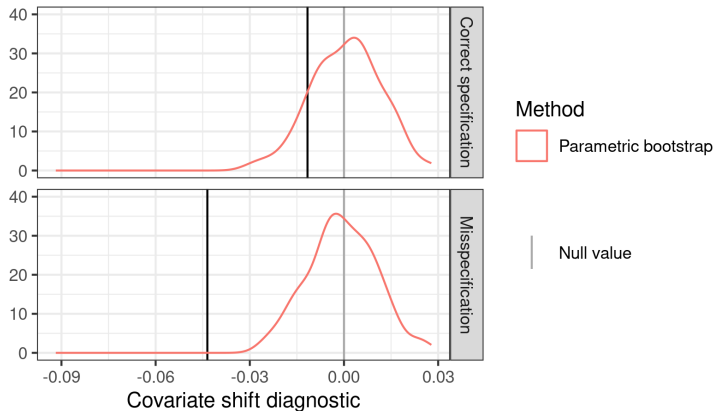
$$\mathcal{P}(\theta|\text{Data}) \propto \exp \left( \sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \text{ and } \mathcal{P}^\epsilon(\theta|\text{Data}) \propto \mathcal{P}(\theta|\text{Data}) \exp \left( \sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta) \right)$$

- $\mathcal{P}(\theta) = \mathcal{P}(\theta|\text{Data})$
- $\phi(\theta) = \frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)$  (so that  $\hat{\mu}^{\text{MrP}}(Y_S) = \mathbb{E}_{\mathcal{P}(\theta|\text{Data})} [\phi(\theta)]$ )
- $h(\theta) = \sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta) \Rightarrow$

Our diagnostic is then given by

$$\Delta(w, \hat{\mathcal{P}}_S) := \left. \frac{\partial \hat{\mu}^{\text{MrP}}(Y_S)}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta|\text{Data})} \left( \frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), \sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta) \right)$$

We expect  $\Delta(w, \hat{\mathcal{P}}_S)$  to be close to zero when the model is correctly specified and posterior estimation is accurate.



**Figure 2:** A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Model 0 (correct) :  $y \sim \text{Logistic}(1 + X_1 + X_2 + X_3 + X_4 + X_5)$

Model 1 (misspecified) :  $y \sim \text{Logistic}(1 + X_3 + X_4 + X_5)$

Compute diagnostic with  $w(x) = \mathbb{I}(X_2 > 0.878)$

Frequentist variance with the infinitesimal  
jackknife

# Why frequentist variance?

For the remainder of the talk, focus on approximating the *frequentist* sampling distribution under  $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$  of quantities like our diagnostic:

$$\Delta = \text{Cov}_{\mathcal{P}(\theta|\text{Data})} \left( \frac{1}{\textcolor{red}{N}_T} \sum_{j=1}^{\textcolor{red}{N}_T} m(\theta^\top x_j), \sum_{i=1}^{\textcolor{blue}{N}_S} w(x_i) \ell(y_i | x_i, \theta) \right)$$

**No need to commit to our covariate shift diagnostic to care!**

- Frequentist variability of *any* model checking diagnostic is interesting
- Frequentist variability remains meaningful under model misspecification (Huggins and Miller, IJ papers)

We will form *local approximations* computable only from *the original posterior*.

We will compare with computationally expensive bootstrap:

- Compute  $\Delta$  on the original dataset
- For  $B = 1, \dots, 100$ :
  1. Sample  $(x_1^*, y_1^*), \dots, (x_{\textcolor{blue}{N}_S}^*, y_{\textcolor{blue}{N}_S}^*)$  (parametric or nonparametric bootstrap)
  2. Run MCMC to estimate  $\mathcal{P}(\theta | (x_1^*, y_1^*), \dots, (x_{\textcolor{blue}{N}_S}^*, y_{\textcolor{blue}{N}_S}^*))$
  3. Use the draws to estimate  $\Delta_B^*$
- Compare  $\Delta$  with quantiles of  $\Delta_B^* - \Delta$



## Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta) \qquad \log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

## Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta) \qquad \log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

Original weights:

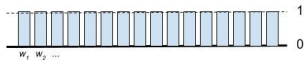


## Data re-weighting.

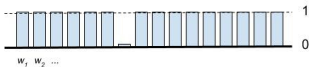
Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta) \qquad \log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

Original weights:



Leave-one-out weights:



# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

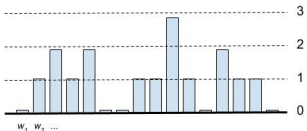
Original weights:



Leave-one-out weights:



Bootstrap weights:



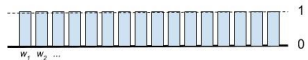
# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

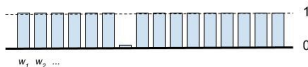
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

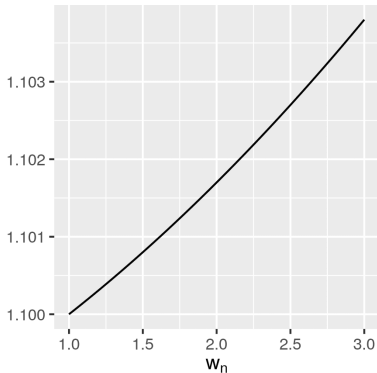
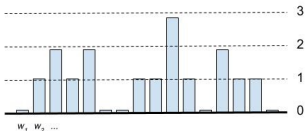
Original weights:



Leave-one-out weights:



Bootstrap weights:



# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

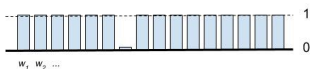
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

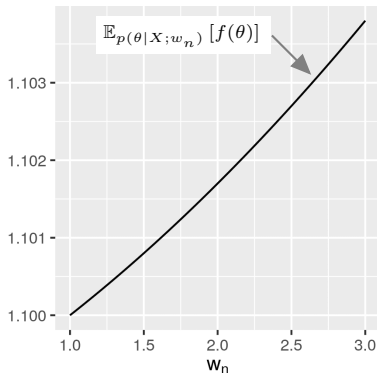
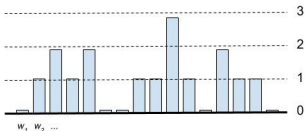
Original weights:



Leave-one-out weights:



Bootstrap weights:



# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

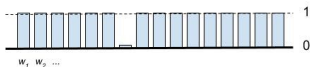
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

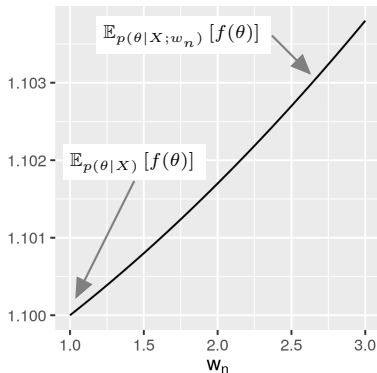
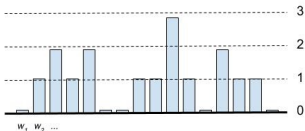
Original weights:



Leave-one-out weights:



Bootstrap weights:



# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

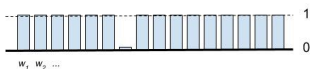
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

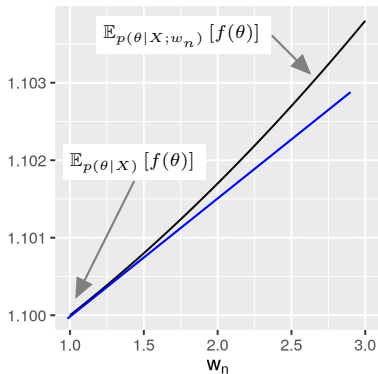
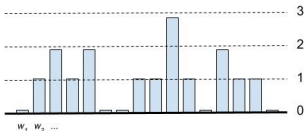
Original weights:



Leave-one-out weights:



Bootstrap weights:





# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

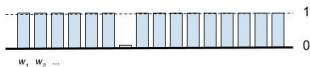
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

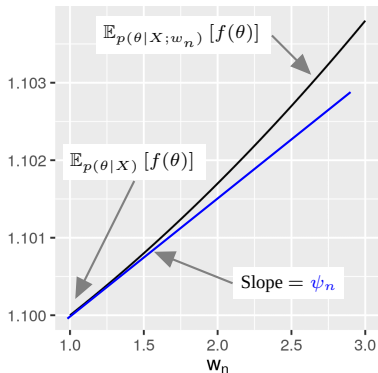
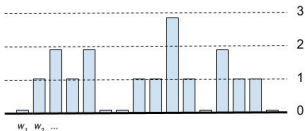
Original weights:



Leave-one-out weights:



Bootstrap weights:



# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

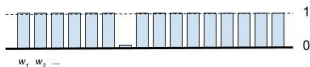
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

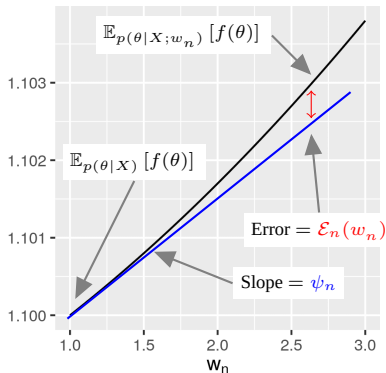
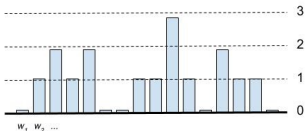
Original weights:



Leave-one-out weights:



Bootstrap weights:



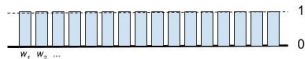
# Data re-weighting.

Augment the problem with *data weights*  $w_1, \dots, w_N$ . We can write  $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$ .

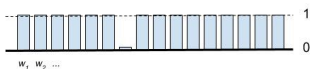
$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta)$$

$$\log \mathcal{P}(X|\theta; w) = \sum_{n=1}^N w_n \ell_n(\theta)$$

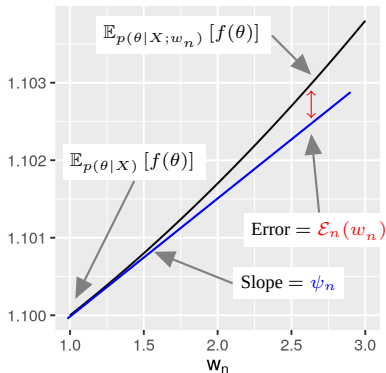
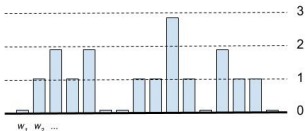
Original weights:



Leave-one-out weights:



Bootstrap weights:



The re-scaled slope  $N\psi_n$  is known as the “influence function” at data point  $x_n$ .

$$\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)] - \mathbb{E}_{\mathcal{P}(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n (w_n - 1) + \mathcal{E}_n(w)$$

How can we use the approximation?

Assume the **slope** is computable and **error** is small.

$$\mathbb{E}_{p(\theta|X;w)} [f(\theta)] - \mathbb{E}_{p(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w)$$

How can we use the approximation?

Assume the **slope** is computable and **error** is small.

$$\mathbb{E}_{p(\theta|X;w)} [f(\theta)] - \mathbb{E}_{p(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w)$$

**Bootstrap.** Draw bootstrap weights  $w \sim \mathcal{P}(w) = \text{Multinomial}(N, N^{-1})$ .

$$\text{Bootstrap variance} = \text{Var}_{\mathcal{P}(w)} \left( \mathbb{E}_{p(\theta|X;w)} [f(\theta)] \right)$$

How can we use the approximation?

Assume the **slope** is computable and **error** is small.

$$\mathbb{E}_{p(\theta|X;w)} [f(\theta)] - \mathbb{E}_{p(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w)$$

**Bootstrap.** Draw bootstrap weights  $w \sim \mathcal{P}(w) = \text{Multinomial}(N, N^{-1})$ .

$$\begin{aligned} \text{Bootstrap variance} &= \text{Var}_{\mathcal{P}(w)} \left( \mathbb{E}_{p(\theta|X;w)} [f(\theta)] \right) \\ &= \text{Var}_{\mathcal{P}(w)} \left( \sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w) \right) \\ &= \frac{1}{N^2} \sum_{n=1}^N \left( \psi_n - \bar{\psi} \right)^2 + \text{Term involving } \mathcal{E}_n(w) \text{ for } n = 1, \dots, N \\ &\approx \frac{1}{N} \underbrace{\left( \frac{1}{N} \sum_{n=1}^N \left( \psi_n - \bar{\psi} \right)^2 \right)}_{\text{"Infinitesimal jackknife variance estimate"}} \end{aligned}$$

# Bayesian von-Mises Expansion

Define the “generalized posterior” functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int g(\theta) \exp \left( N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left( N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Recall that  $\mathbb{E}_{\mathcal{P}(\theta|\text{Data})} [g(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, study the *von Mises expansion*.

Let  $\mathcal{P}_t = t\hat{\mathcal{P}}_S + (1 - t)\mathcal{P}_S$ . Then form the series expansion in  $t$ :

$$\begin{aligned} \sqrt{N_S} \left( \mathbb{E}_{\mathcal{P}(\theta|\text{Data})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) &= \sqrt{N_S} \left( T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S) \right) \\ &= \sqrt{N_S} \left. \frac{\partial T(\mathcal{P}_t, N_S)}{\partial t} \right|_{t=0} (1 - 0) + \mathcal{E}(\tilde{t}) \\ &= \underbrace{\sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

where

$$\psi_i := N_S \text{Cov}_{\mathcal{P}(\theta|\text{Data})} (g(\theta), \ell(y_i|x_i)) \quad \text{and} \quad \tilde{t} \in [0, 1].$$

# Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left( \mathbb{E}_{\mathcal{P}(\theta|\text{Data})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

- Assume (sketch): A well-behaved MAP *maximum a posteriori* estimator  $\hat{\theta}$  exists.
  - The dimension of  $\theta$  is fixed as  $N \rightarrow \infty$
  - The expected log likelihood has a strict maximum at  $\theta_0$
  - The observed log likelihood satisfies  $\hat{\theta} \rightarrow \theta_0$
  - The expected log likelihood Hessian is negative definite at  $\theta_0$
- Assume (sketch): We can apply standard asymptotics.
  - The log prior and log likelihood are four times continuously differentiable
  - The prior is proper, and a technical set of prior expectations are finite
  - The log likelihood and its derivatives are dominated by a square-integrable envelope function for all  $\theta$  in a neighborhood of  $\theta_\infty$ .

## Theorem 3 [Giordano and Broderick, 2026] (sketch):

$$\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0 \quad \text{in the Bayesian von-Mises expansion.}$$

**Corollary:** Let  $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$ . By the CLT applied to  $\psi_i$ ,

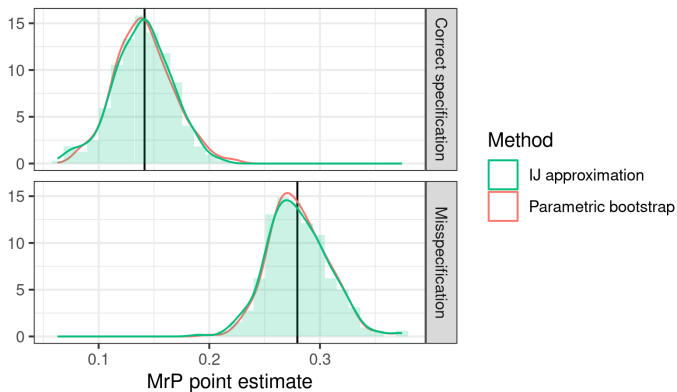
$$\sqrt{N_S} \left( \mathbb{E}_{\mathcal{P}(\theta|\text{Data})} [g(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$



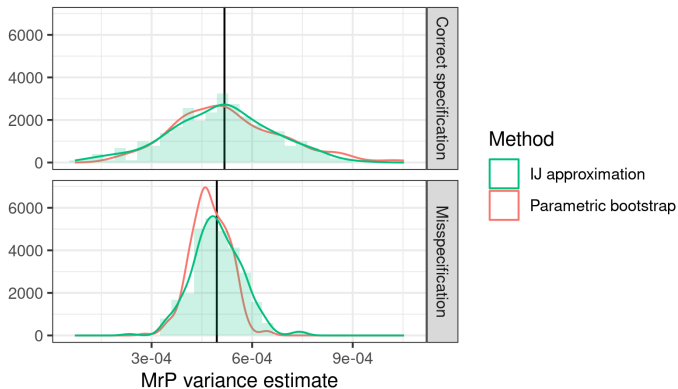
Recall that our covariate shift diagnostic is given by

$$\Delta(w, \hat{\mathcal{P}}_S) = \text{Cov}_{\mathcal{P}(\theta|\text{Data})} \left( \frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), \sum_{i=1}^{N_S} w(x_i) \ell(y_i | x_i, \theta) \right) = \sum_{i=1}^{N_S} \psi_i w(x_i).$$

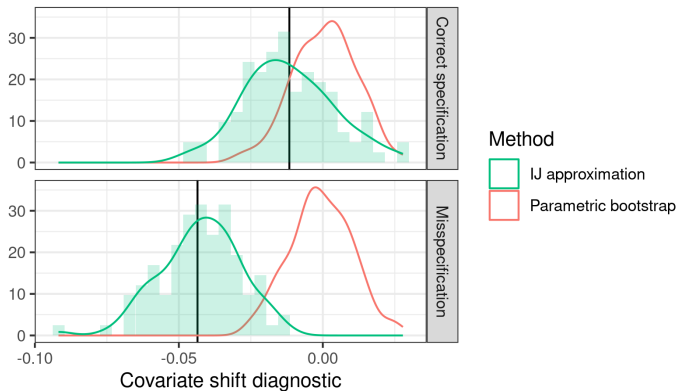
In other words, we are checking whether the influence function aligns with any of the regressors.



**Figure 3:** MrP Posterior Mean



**Figure 3:** MrP Posterior Variance



**Figure 3:** Covariate Shift Diagnostic

Local robustness with flows

## Why flows for sensitivity analysis?

The IJ is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [g(\theta)] \approx \mathbb{E}_{\mathcal{P}} [g(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [g(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws.

This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantities are hard to linearize
- Bayesians like to work with draws, not analytic expectations

**Idea:** Instead of linearizing expectations, learn a flow from  $\mathcal{P}$  to  $\mathcal{P}^\epsilon$ .

There's a large literature on learning flows between distant distributions. Our problem is easier:

- Only need “local” flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Consider a family of distributions  $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$  and a test function  $\phi(\theta)$ . Then, by interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

We want a flow  $\theta \mapsto f^\epsilon(\theta)$  such that, for small  $\epsilon$ ,  $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$  and  $f^0(\theta) = \theta$ .

Suppose there is a scalar-valued potential function  $v(\theta)$  such that

$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2)$  as  $\epsilon \rightarrow 0$ . Then:

$$\begin{aligned} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] &= \mathbb{E}_{\mathcal{P}(\theta)} [\phi(f^\epsilon(\theta))] \Rightarrow \\ \left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] &= \left. \frac{\partial}{\partial \epsilon} \right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta + \epsilon \nabla v(\theta))] \\ \parallel &\parallel \\ \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta) h(\theta)] &= \mathbb{E}_{\mathcal{P}(\theta)} [\nabla \phi(\theta)^\top \nabla v(\theta)] \end{aligned}$$

- When does such a  $v(\theta)$  exist?
- How can we find it in practice?

## When does such a $v(\theta)$ exist?

Consider a family of distributions  $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$  and a test function  $\phi(\theta)$ . Then, by interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

We want a flow  $\theta \mapsto f^\epsilon(\theta)$  such that, for small  $\epsilon$ ,  $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$  and  $f^0(\theta) = \theta$ .

### When does such a $v(\theta)$ exist?

**Theorem (Schewengber 2026):** Assume that

- $\log \mathcal{P}(\theta) \in C^2(\mathcal{R}^d)$
- $\lim_{\theta \rightarrow \infty} \left( \frac{1}{2} \|\nabla \log \mathcal{P}(\theta)\|^2 - \Delta \log \mathcal{P}(\theta) \right) = +\infty$ .
- $\mathbb{E}_{\mathcal{P}(\theta)} [h(\theta)^2] < \infty$

Then there exists a solution  $v$  to the differential equation

$$-\mathcal{L}_{\mathcal{P}}(v) := \nabla v^\top \nabla \log \mathcal{P} + \Delta v = h$$

such that  $\|\nabla v\|_{L^2(\mathcal{P})} < \infty$ , and the flow  $\theta_\epsilon = \theta + \epsilon \nabla v(\theta)$  satisfies  $\theta_\epsilon \sim \mathcal{P}^\epsilon(\theta)$  for small  $\epsilon$ .

Note that  $\mathcal{L}_{\mathcal{P}}$  is the drift operator, generator of Langevin diffusion, trace of the Stein operator.



## How can we find $v(\theta)$ in practice?

Fix  $h(\theta)$ . Specify

- A family of real-valued test functions  $\phi_n(\theta)$  for  $n \in [N]$
- A family of real-valued basis functions  $b_m(\theta)$  for  $m \in [M]$

Let  $v(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$ . Then

$$\underbrace{\mathbb{E}_{\mathcal{P}(\theta)} [\phi_n(\theta) h(\theta)]}_{\text{Estimate with MCMC!}} = \sum_{m=1}^M \beta_m \underbrace{\mathbb{E}_{\mathcal{P}(\theta)} [\nabla \phi_n(\theta)^\top \nabla b_m(\theta)]}_{\text{Estimate with MCMC!}}$$

This gives  $N$  linear equations in  $M$  unknowns. Known as “Otto’s calculus” in differential equations. Formally similar to numerical PDE solutions known as Galerkin methods, except we have a stochastic grid ( $\theta \sim \mathcal{P}(\theta)$ ) rather than a deterministic one.

Unlike standard linear regression:

- We are free to choose  $\phi_n$  and  $b_m$
- Both  $F$  and  $B$  are measured with noise
- We have a lot of information about the covariance of  $F$  and  $B$

**Theorem (Schewengber 2026, sketch):** The solution gives the best L2 approximation to  $\|\nabla v(\theta)\|_{L^2(\mathcal{P})}$  in the span of  $\phi_n(\theta)$ . When  $\mathcal{P}$  is Gaussian, the ordered Hermite polynomials give the best finite-rank approximation.

Suppose we believe that  $\mathcal{P}(\theta|\text{Data})$  is approximately Gaussian.

Take the basis functions  $b_m(\theta) = \theta_m$  (1st Hermite polynomial) and test functions  $\phi_n(\theta) = b_m(\theta)$ . Then the optimal flow is given by

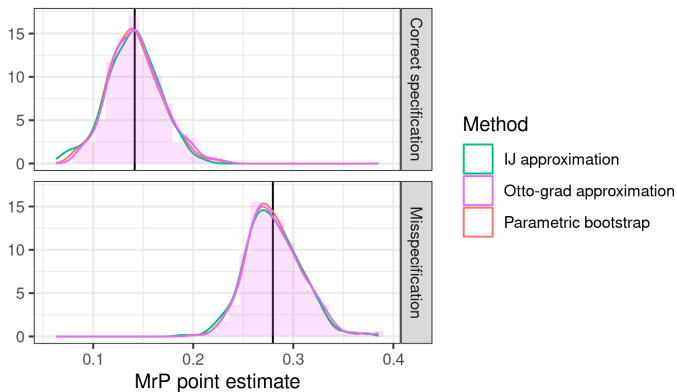
$$\theta_\epsilon = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\text{Data})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\text{Data})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

**This just translates the posterior in the direction of the mean changes.**

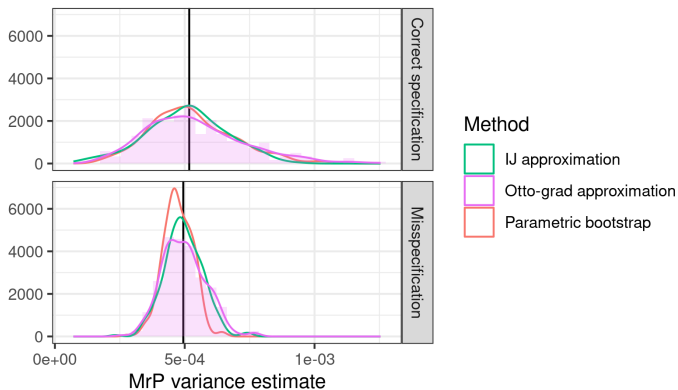
Suppose we have  $\theta_m \sim \mathcal{P}(\theta|\text{Data})$ , and bootstrap weights  $w_i$ . Set

$$\theta_m^* = \theta_m + \frac{1}{N_S} \sum_{i=1}^{N_S} w_i \psi_i$$

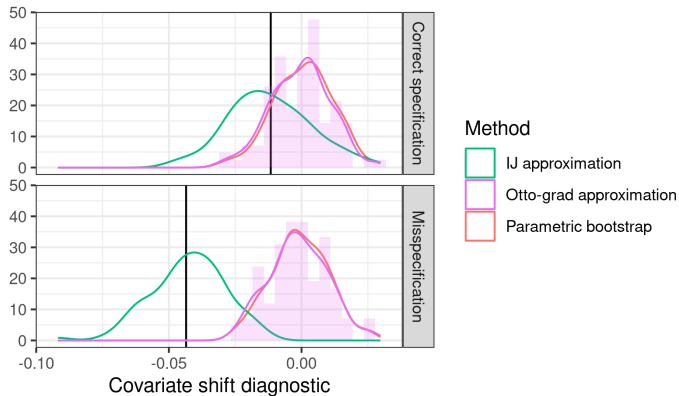
and compute any posterior quantities from these samples.



**Figure 4:** MrP Posterior Mean



**Figure 4:** MrP Posterior Variance



**Figure 4:** Covariate Shift Diagnostic

R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2026.